

Corporation of the City of Quinte West

Trenton/Bayside Drinking Water System (Trenton Service Area)

2019 Annual Drinking Water System Report



A Natural Attraction



The Corporation of the City of Quinte West
Public Works and Environmental Services
Water/Wastewater Division
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Drinking Water System Information

Drinking Water System Number:	220001619
Drinking Water System Name:	Trenton/Bayside Drinking Water System (Trenton Service Area)
Drinking Water System Owner:	The Corporation of the City of Quinte West
Drinking Water System Category:	Large Municipal-Residential System
Period being Reported:	January 1, 2019 through December 31, 2019

Does your Drinking Water System service more than 10,000 people?

Yes

Is your Annual Report available to the public at no charge on a website

Yes, please visit www.quintewest.ca

Location where Summary Report required under O.Reg.170/03 Schedule 22 will be available for inspection

Water/Wastewater Division Administration Office

25 Couch Crescent

Trenton, ON, K8V 1G8

List all Drinking Water Systems (if any), that receive all of their Drinking Water from your System:

Carrying Place/Consecon Water Distribution System, DWS number 260005099



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Indicate how you notified system users that your annual report is available, and is free of charge.

- ☒ Public access/notice via City website
- ☐ Public access/notice via Government Office
- ☒ Public access/notice via newspaper
- ☒ Public access/notice via City social media platform(s)
- ☐ Public access/notice via a Public Library
- ☐ Public access/notice via other method:

Description of the Drinking Water System

The Trenton Water Treatment Plant draws water from the Trent River upstream of Dam No. 1 through two intake pipes; one 53 m long, 400 mm diameter raw water intake pipe, and a second, 18 m long, 600 mm diameter intake. This conventional, chemically assisted filtration plant has a rated capacity of 35,800 cu.m/day. Processes used at the filtration plant include flocculation, sedimentation, and Granular Activated Carbon Filtration. Chlorine gas is applied as a disinfectant before filtered water enters two interconnected baffled clearwells with a combined capacity of 5,454 cu.m. The potable (drinkable) water is then pumped into the distribution system through a set of four (4) highlift pumps. There are two elevated water storage tanks in service. The first is the 2nd Dug Hill Road Water Tower; capacity 2,273 cu.m, the second is the Oak Street Water Tower; capacity 2,273 cu.m. Three Booster Pumping Stations are located within the Trenton WDS; they are Catherine Street Booster Pumping Station, Mount Pelion Booster Pumping Station, and Telephone Road Booster Pumping Station. These stations are used to boost system pressure, and supply water into areas at a higher elevation than the Water Treatment Plant.

The Trenton distribution system services approximately 22,000 people in the Trenton community, in addition to part of CFB Trenton. The Trenton Water Treatment plant also supplies all drinking water to the Carrying Place and Consecon Water Distribution System in



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Prince Edward County. In 2015 the City installed a watermain along Old Hwy 2 that effectively connected the Trenton Water Distribution System (WDS) to the Bayside WDS. On April 21, 2017 the Trenton, and Bayside Drinking Water System's were consolidated and governed under one Drinking Water Works Permit (DWWP) # 163-202, and Municipal Drinking Water Licence (MDWL) # 163-102. Each system is still assigned a separate Drinking-Water System number by the Ministry of Environment, Conservation and Parks.

List all water treatment chemical used over this Reporting Period:

- ✓ Aluminum Sulphate (Tradename: Alum)
- ✓ Chlorine Gas

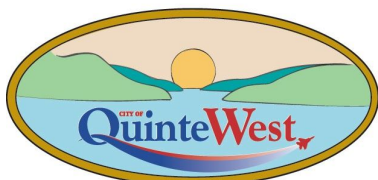
Were any significant expenses incurred to:

- ✓ Install Required Equipment?
- ☐ Repair required Equipment?
- ✓ Replace required Equipment?

Provide a brief description, and a breakdown of monetary expenses incurred to facilitate Equipment upgrades:

Preventative Maintenance (PM) activities for lubrication, inspections, testing and cleaning of equipment is scheduled and completed routinely, along with other lifecycle replacement needs. In addition to the PM activity, the following Capital expenditures were incurred this Reporting Period:

Replace six (6) rotork actuators	~ \$30,000
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Replace Mount Pelion communication tower	~\$177,000
Install dechlorination equipment and contact piping for waste tank decant system	~ \$60,000
Water Meter automation program	~ \$148,000
Construct new watermain to service new Industrial Park	~ \$161,000
Replace watermain on South St, Heber St, Crown Street, George Street	\$603,000

Notices submitted in accordance with subsection 18(1) of the Safe Drinking Water Act or section 16-4 of Schedule 16 of O.Reg. 170/03 and reported to Spills Action Centre during this Reporting Period:

None.

Microbiological Testing completed in accordance with Schedule 10, 11, or 12 of Regulation 170/03 during this Reporting Period:

	Number of Samples	Range of E.Coli (min - max)	Range of Total Coliform (min - max)	Number of Heterotrophic Plate Count Samples	Range of HPC results (min - max)
Raw	53	4 - 760	22 - 18400	NA	NA
Treated	53	0 - 0	0 - 0	53	0 - 1
Distribution	424	0 - 0	0 - 0	216	0 - 6



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**Operational Testing completed in accordance with
Schedule 7, 8, or 9 of O.Reg.170/03 during this Reporting
Period:**

Parameter	Number of Grab Samples	Range of Results (min-max)
Turbidity, (NTU)	8760	0.000 ¹ - 0.634
Primary Disinfection FAC ² , (mg/L)	8760	1.54 - 2.79
Secondary Disinfection FAC, (mg/L)	8760	0.49 - 2.85

**Summary of additional testing and sampling carried out in
accordance with the requirement of an Approval, Order,
or other Legal Instrument:**

In accordance with Condition 4.0 of MDWL #163-102, monthly samples must be collected and tested for Total Suspended Solids, and Total Residual Chlorine, from the following waste streams that may impact the natural environment:

- Waste Residual Management Sedimentation Tank - Clear Water Discharge

Condition 1.5 of the MDWL outlines the maximum annual average concentration for each of these parameters. Due to detectable TRC³ residual in cold water temperatures in 2018, the decant system had to be shut down and a dechlorination system installed.

¹ Values occur on analyzer failure throughout the reporting period. Three (3) events recorded with no regulatory requirements.

² Free Available Chlorine. Defined as the free amount of chlorine available in water.

³ Total Residual Chlorine. Defined as the total amount of chlorine available in water.



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No environmental discharge was facilitated throughout the reporting period.

Summary of Inorganic parameters tested during this Reporting Period or the most recent Sample Results:

Inorganic Compound Results				
Parameter	Sample Date (dd/mmm/yy)	Result Value	Unit of Measure	Exceedance?
Antimony	02-JAN-19	0.23	ug/L	NO
Arsenic	02-JAN-19	0.2	ug/L	NO
Barium	02-JAN-19	29.9	ug/L	NO
Boron	02-JAN-19	8	ug/L	NO
Cadmium	02-JAN-19	0.003	ug/L	NO
Chromium	02-JAN-19	0.08	ug/L	NO
Mercury	02-JAN-19	0.01	ug/L	NO
Selenium	02-JAN-19	0.06	ug/L	NO
Sodium	02-JAN-19	9.87	mg/L	NO
Fluoride	02-JAN-19	0.06	mg/L	NO
Uranium	02-JAN-19	0.019	ug/L	NO
Nitrate	02-JAN-19 02-APR-19 03-JUL-19 01-OCT-19	0.344 0.416 0.153 0.061	mg/L	NO
Nitrite	02-JAN-19 02-APR-19 03-JUL-19 01-OCT-19	0.003 0.003 0.003 0.003	mg/L	NO



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Summary of Lead Testing under Schedule 15.1 during this Reporting Period:

Lead Sampling Results⁴				
Sampling Period	Location Type	Number of Samples	Range of Lead Results (ug/L), 'min-max'	Number of Exceedances
Winter Sampling Period (09-APR-19)	Distribution	4	0.02 - 0.24	0
Summer Sampling Period (16-SEP-19)	Distribution	4	0.01 - 0.33	0

Summary of Organic parameters sampled during this Reporting Period:

Organic Compound Results				
Parameter	Sample Date	Result Value	Unit of Measure	Exceedance
Alachlor	02-JAN-19	0.02	ug/L	No
Atrazine + N-dealkylated metabolites	02-JAN-19	0.01	ug/L	No

⁴ Samples collected in response to customer inquiries fell outside of the Sampling Periods. Three sets of samples collected between December 12&13 measured below the regulatory limit established for Lead.



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Azinphos-methyl	02-JAN-19	0.05	ug/L	No
Benzene	02-JAN-19	0.32	ug/L	No
Benzo(a)pyrene	02-JAN-19	0.004	ug/L	No
Bromoxynil	02-JAN-19	0.33	ug/L	No
Carbaryl	02-JAN-19	0.05	ug/L	No
Carbofuran	02-JAN-19	0.01	ug/L	No
Carbon Tetrachloride	02-JAN-19	0.16	ug/L	No
Chlorpyrifos	02-JAN-19	0.02	ug/L	No
Diazinon	02-JAN-19	0.02	ug/L	No
Dicamba	02-JAN-19	0.20	ug/L	No
1,2-Dichlorobenzene	02-JAN-19	0.41	ug/L	No
1,4-Dichlorobenzene	02-JAN-19	0.36	ug/L	No
1,2-Dichloroethane	02-JAN-19	0.35	ug/L	No
1,1-Dichloroethylene (vinylidene chloride)	02-JAN-19	0.33	ug/L	No
Dichloromethane	02-JAN-19	0.35	ug/L	No



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2-4 Dichlorophenol	02-JAN-19	0.15	ug/L	No
2,4-Dichlorophenoxy acetic acid (2,4-D)	02-JAN-19	0.19	ug/L	No
Diclofop-methyl	02-JAN-19	0.40	ug/L	No
Dimethoate	02-JAN-19	0.03	ug/L	No
Diquat	02-JAN-19	1	ug/L	No
Diuron	02-JAN-19	0.03	ug/L	No
Glyphosate	02-JAN-19	1	ug/L	No
Malathion	02-JAN-19	0.02	ug/L	No
2 methyl-4- chlorophenoxyacetic acid (MCPA)	02-JAN-19	0.00012	mg/L	No
Metolachlor	02-JAN-19	0.01	ug/L	No
Metribuzin	02-JAN-19	0.02	ug/L	No
Monochlorobenzene	02-JAN-19	0.3	ug/L	No
Paraquat	02-JAN-19	1	ug/L	No
Pentachlorophenol	02-JAN-19	0.15	ug/L	No



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Phorate	02-JAN-19	0.01	ug/L	No
Picloram	02-JAN-19	1	ug/L	No
Polychlorinated Biphenyls(PCB)	02-JAN-19	0.04	ug/L	No
Prometryne	02-JAN-19	0.03	ug/L	No
Simazine	02-JAN-19	0.01	ug/L	No
THM (NOTE: show latest annual average)	01-OCT-19	61.75	ug/L	No
Terbufos	02-JAN-19	0.01	ug/L	No
Tetrachloroethylene	02-JAN-19	0.35	ug/L	No
2,3,4,6-Tetrachlorophenol	02-JAN-19	0.20	ug/L	No
Triallate	02-JAN-19	0.01	ug/L	No
Trichloroethylene	02-JAN-19	0.44	ug/L	No
2,4,6-Trichlorophenol	02-JAN-19	0.25	ug/L	No
Trifluralin	02-JAN-19	0.02	ug/L	No
Vinyl Chloride	02-JAN-19	0.17	ug/L	No



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**Inorganic or Organic parameter(s) that have exceeded
half the standard prescribed in Schedule 2 of Ontario
Drinking Water Quality Standards:**

None.



2019 Summary Report to Council

In accordance with Schedule 22 requirements outlined in Ontario Regulation 170/03, a Summary Report shall be prepared no later than March 31 for the preceding year, and supplied to members of municipal council.

The Report shall list the requirements of the Act, Regulations, Drinking Water Works Permit, Municipal Drinking Water Licence, and any Orders applicable to the system that were not met at any time during the period covered by the Report.

The Report must also include a summary of the quantities and flow rates of potable (drinkable) water supplied during the Reporting Period, including monthly average and maximum daily flows. A comparison of these flows to the rated capacity and flow rates approved in the system Drinking Water Works Permit, and Municipal Drinking Water Licence, must also be provided.

Prescribed Instruments applicable to the Trenton DWS

The Trenton DWS is governed by, and must operate their DWS primarily in accordance with, the following Acts and Regulations at minimum:

- Safe Drinking Water Act, 2002;
- O. Reg. 128/04 – Certification of Drinking Water System Operators and WQA
- O. Reg. 170/03 – Drinking Water Systems
- O. Reg. 169/03 – Ontario Drinking Water Quality Standards
- Environmental Protection Act, where applicable;
- Clean Water Act, where applicable;
- Municipal Drinking Water Licence 163-102;
- Drinking Water Works Permit 163-202;
- Permit to Take Water 1007-9HJP6L.



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Councillors wishing to obtain a copy of any Act or Regulation may contact the Compliance Coordinator at amyr@quintewest.ca to obtain a current consolidated copy and interpretation of the legislation.

Compliance with Prescribed Instruments, Acts and Regulations

Safe Drinking Water Act

NSF International conducted an onsite Re-Accreditation Audit between November 12, 2019 and November 14, 2019. The intent of the onsite audit is to confirm the City's conformance to the Drinking Water Quality Management Standard. The City's registration has been renewed for another three (3) year period and will expire July 23, 2022 as a result of a successful audit.

Clean Water Act

The Source Protection Plan was approved by the Minister of Environment, Conservation and Parks, and came into effect on January 1, 2015. The City has put the necessary internal processes in place with the Planning, Building, and Public Works Departments to screen applications and ensure compliance with the Source Protection Plan. In 2018 there were 17 applications submitted to the Risk Management Official for review and approval. In general the screening, application, and approval processes implemented by staff seem to be working reasonably well. City staff will begin working with Lower Trent Conservation Authority in 2020 to develop Risk Management Plans with applicable property owners.



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Permit to Take Water

The City operates its Trenton Water Treatment Plant in accordance with Permit to Take Water (PTTW) number 1007-9JHP6L which expires on April 30, 2024. This Permit allows the City to take 35,800 cu.m/day from the Trent River watershed at a maximum flow rate of 530.4 L/s. The maximum recorded daily taking was 15,533 cu.m/day on July 9, 2019. The maximum recorded flow rate was 473.4 L/s on June 27, 2019.

For a detailed summary of water quantities and flow rates, see [2019 Water Quantities and Flow Rates - Raw Water Assessment](#) on page 18

Drinking Water Works Permit/Municipal Drinking Water Licence

City staff are preparing to submit an application for renewal of its Municipal Drinking Water Licence and subsequent Drinking Water Works Permit which is due to expire on October 18, 2020. With the application, the City is required to submit an updated 10-year Financial Plan, updated Water Distribution System mapping, and proof of successful NSF accreditation.

In accordance with Condition 1.1 of the MDWL, the Trenton Water Treatment Plant has a rated capacity to treat and distribute 35,800 cu.m/day. Over the Reporting Period the maximum daily volume of treated water distributed to the distribution system was 14,591 cu.m/day on July 9, 2019. The Annual Average Daily Flow pumped to the Distribution System was 9,334 cu.m/day. For a detailed summary of treated water quantities and flow rates, see [2019 Water Quantities and Flow Rates - Treated Water Assessment](#) on page 19.



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2019 Water Quantity and Flow Rates (Raw Water Assessment)

Raw Water Flow: PTTW Limit = 35,800 cu.m/d			
Month	Average Daily Flow (cu.m/d)	Maximum Daily Flow (cu.m/d)	Total Monthly Flow (cu.m/month)
January	9,341	10,193	289,565
February	9,345	10,066	261,653
March	9,598	10,406	297,523
April	9,655	10,739	289,642
May	10,124	11,225	313,834
June	10,880	13,278	326,412
July	12,739	15,533	394,910
August	11,710	13,007	363,019
September	10,670	12,000	320,109
October	9,573	11,557	296,763
November	9,501	15,231	285,038
December	9,005	10,175	279,169
Total Raw Water Flow =			3,717,637



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2019 Water Quantity and Flow Rates (Treated Water Assessment)

Treated Water Flow: Facility Rated Capacity = 35,800 cu.m/day			
Month	Average Daily Flow (cu.m/d)	Maximum Daily Flow (cu.m/d)	Total Monthly Flow (cu.m/month)
January	8,536	9,337	264,630
February	8,485	9,247	237,571
March	8,682	9,420	269,137
April	8,675	9,515	260,243
May	9,248	10,357	286,692
June	10,061	12,203	301,843
July	11,861	14,591	367,681
August	10,838	12,079	335,991
September	9,882	11,135	296,471
October	8,798	10,569	272,748
November	8,670	14,431	260,094
December	8,184	9,257	253,700
Total Treated Water Flow =			3,406,800

- Annual Average Daily (Treated Water) Flow = 9,334 cu.m/day. This accounts for 26% of the facility Rated Capacity.
- Maximum Daily (Treated Water) Flow = 14,591 cu.m/day. This accounts for 41% of the facility Rated Capacity during record peak daily flow conditions.



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Historical Flow Comparison

Flows have been quite stable since 2017 when a significant repair was facilitated in the water distribution system. Continual Capital replacement of aging Infrastructure may be contributable to the stability in flows over the last few years as well, considering ongoing population growth in the community. The 5-year AADF is 10,092 cu.m/day which accounts for 28% of the Rated Capacity. This indicates there is adequate Reserve Capacity in the facility operation for future growth.

